Low Voltage AC Circuit Breaker

Lecture 8

Definition

- The circuit breakers that are used for breaking and making circuits below 1000 volts are called low voltage circuit breakers. The definition of low voltages depends on its context being used for. According to IEC, low voltage refers to voltage below 1000v. Arc generated at such voltages are easily extinguishable. LV circuit breakers are mostly used for residential and industrial applications.

Low voltage AC circuit breaker types:

- Miniature Circuit Breaker (MCB)
- Molded Case Circuit Breaker (MCCB)
- Residual Current Circuit Breaker (RCCB)
- GFI or GFCI circuit breaker (Ground fault circuit interrupter)
- Arc Fault circuit interrupter (AFCI)
- Common trip (ganged) breakers.
- Magnetic circuit breakers.
- Thermal magnetic circuit breakers.

Miniature Circuit Breaker (MCB)

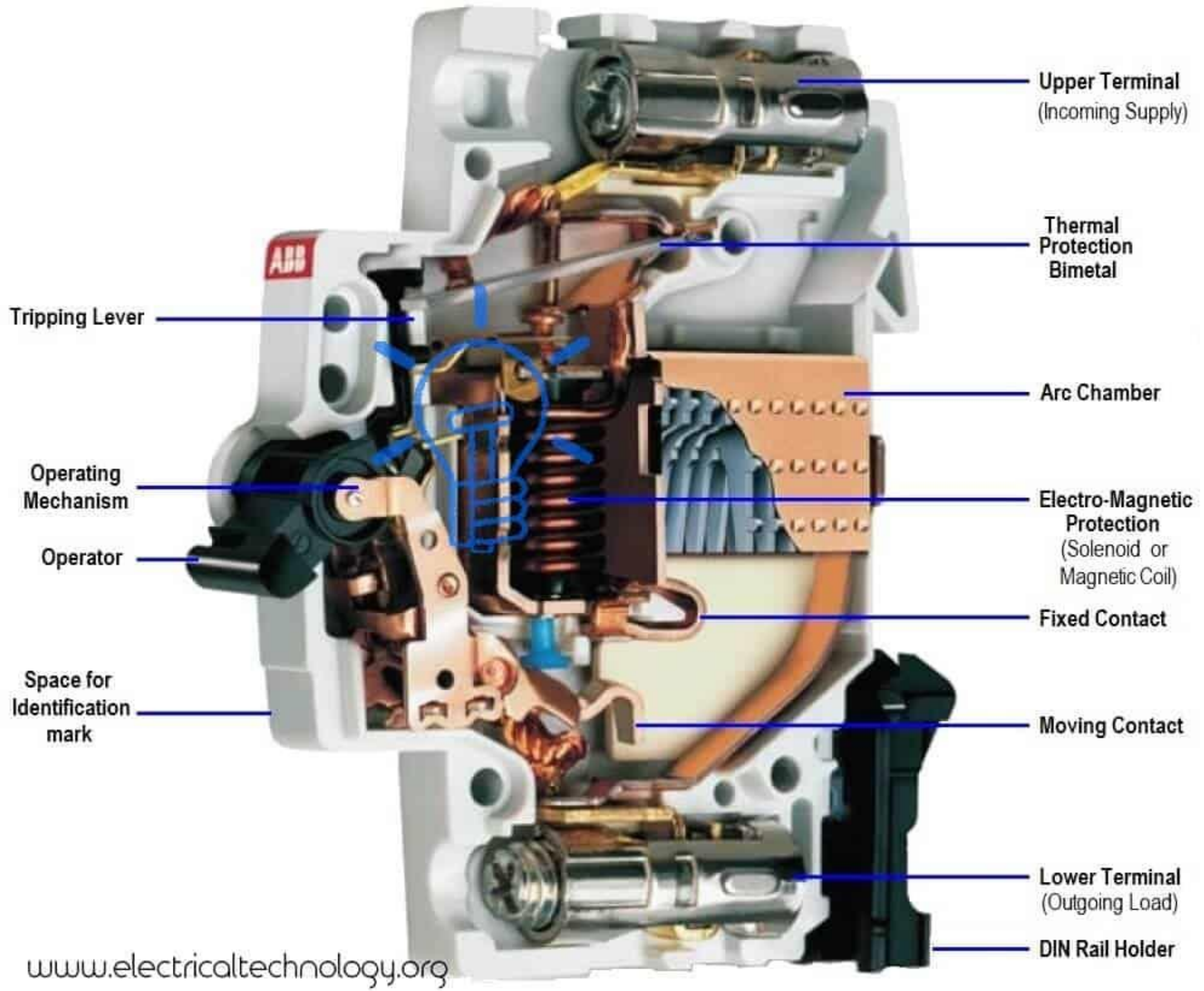
- **MCB or Miniature Circuit Breaker** is an automatically operated electromechanical device used for the protection of the circuit from overloading or short circuit. It breaks or opens the circuit when the current flowing through it exceeds its rated limit. MCB is used for the protection of low voltage circuit 240/415 v AC having a wide range of current ratings below 125V.
- MCB doesn’t trip (switch off) instantly; instead, there is a time delay between fault occurrence and the breaking of contacts. Generally, they are designed to have a time delay of less than 2.5 millisecond for short circuit and 2 sec to 2 min for overloading. It is to make sure the CB does not trip every time with a momentary surge or starting of inductive load due to high inrush current from such loads such as electrical motors.

MCB does not have adjustable trip characteristics. The breaking mechanism could be either thermal or thermal-magnetic in operation. The thermal breaking mechanism is used in case of overloading while the magnetic breaking mechanism is used in case of a short circuit.

MCB is enclosed in an insulating casing. The fixed and moving contacts made of copper or silver alloy connect with the two terminals for current supply. There is an arc chute consisting of multiple conducting plates called arc splitters that dissipate the arc energy. While the operating mechanism, is of two types i.e. thermal and magnetic.

The thermal tripping mechanism consists of a bimetallic strip (made from two different metals having different thermal expansion) usually made from steel and brass is used for breaking the circuit in case of overloading. When the current above-rated limit starts flowing through the metallic strip, it heats up and starts expanding due to which it bents and triggers the latch to separate the contacts.

The magnetic tripping mechanism consists of a coil or solenoid that produces a magnetic field when current flows through it. In case of a short circuit or very high current, the solenoid produces a strong magnetic field to pull the lever and separate the contacts.



Upper Terminal
(Incoming Supply)
Thermal Protection Bimetal
Tripping Lever
Arc Chamber
Operating Mechanism
Electro-Magnetic Protection
(Solenoid or Magnetic Coil)
Operator
Fixed Contact
Space for Identification mark
Moving Contact
Lower Terminal
(Outgoing Load)
DIN Rail Holder
ABB
www.electricaltechnology.org

Molded Case Circuit Breaker (MCCB)

- **MCCB or Molded Case Circuit breaker** is an electromechanical circuit breaker having very high current ratings up to 2500 Amps. It is used in applications where the current ratings exceed the range of MCB (Miniature Circuit Breaker).
- It offers a thermal-magnetic tripping mechanism where the thermal mechanism is used for overloading and magnetic is used for short circuit conditions. It can interrupt current around 10k – 200k amps.
- MCCB trip characteristics are adjustable in any current rating. MCB does not have such a feature. MCCB is suitable for applications where normal current is above 100 amps. They are installed in industries.

MCCB can have a fixed or interchangeable trip unit. The trip unit is responsible for breaking the contacts upon meeting the fault condition. It provides three types of functions:

Overloading: Overloading occurs when the current exceeds a certain limit for a specific duration. Such current can damage the equipment and wiring and cause fire hazards. MCCB uses a bimetallic strip to protect against overloading.

The bimetallic strip is made of two types of metal having a different rate of heat expansion. The current flowing is used for heating the strip by either using a heater coil or directly conducting it through it. The strip heats up and bents thus tripping the mechanism.

Short Circuit: Short circuit current refers to the fault current in the system due to a downed line or broken exposed wires that come in contact or faulty equipment. the current flow due to short circuit is very large far more than the overloading current.

Short circuit must be interrupted in the shortest period of time possible. MCCB can trip SC currents up to 10K-200K amps in the duration of 0.04 seconds.

Manual Switching

MCCB can also act as a manual switch to switch ON/OFF the power supply to the circuit. It can manually break the power supply in case of emergency or maintenance.

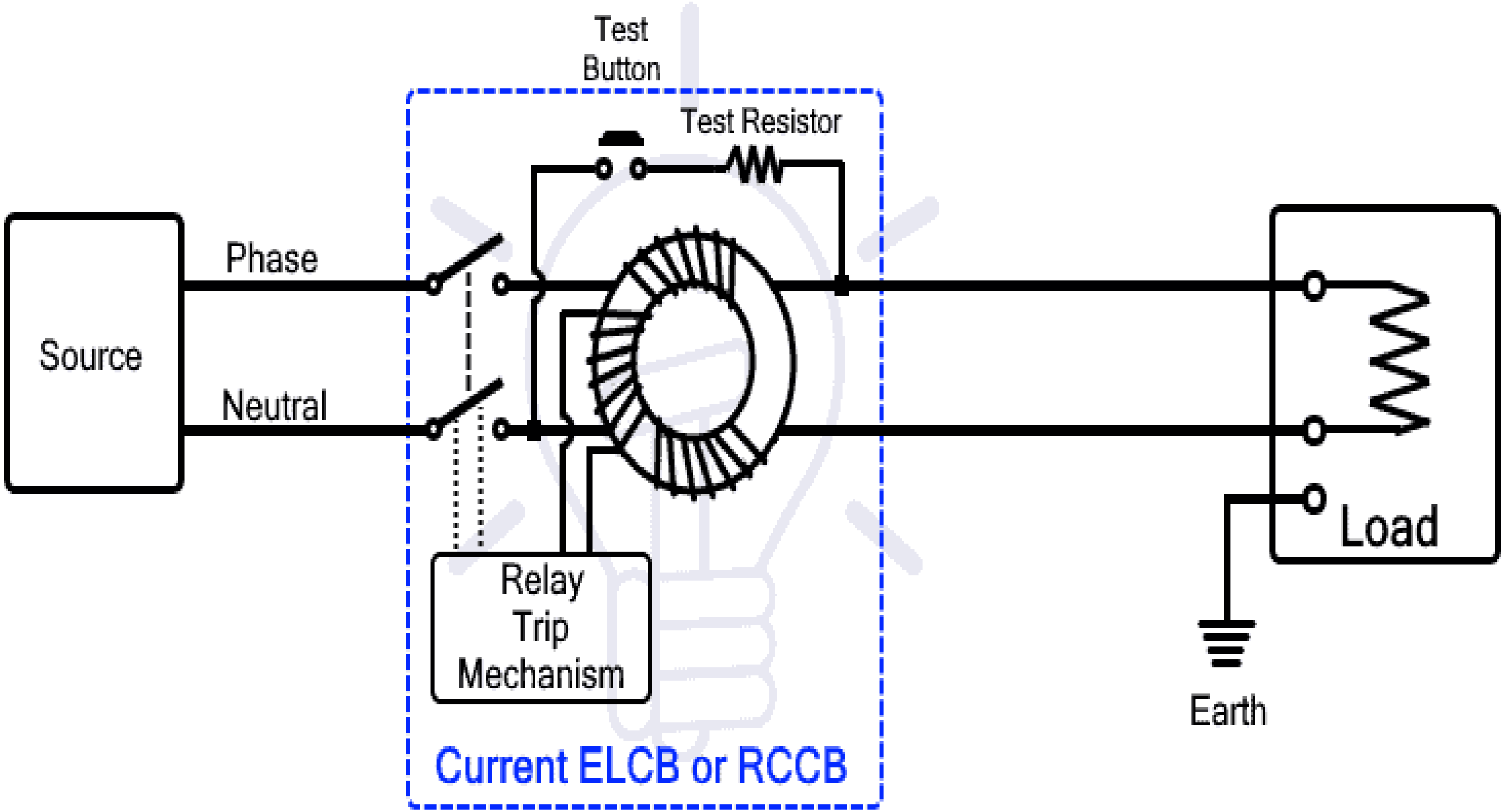
Advantages of MCCB

- MCCB has an adjustable trip setting.
- It can interrupt very high currents.
- It has a movable trip unit.
- It has a very small tripping time thus fast switching during fault current.
- It also offers remote ON/OFF feature.
- It has a compact design and takes less space

Residual Current Circuit Breaker (RCCB)

- **The current Earth Leakage Circuit Breaker ELCB is generally known as RCD or RCCB.** Residual Current Device (RCD) or Residual Current Circuit Breaker (RCCB) is a type of ELCB that breaks the circuit in case of leakage current. It helps in protection against electrical shock.
- The current leakage occurs when the current flows in an accidental path. In normal conditions, the current flows into the load through a hot or live wire and flows out of the load through the neutral wire.

The current leaks if the current flows out through the ground wire or through a person's body connected with the ground.
RCCB works on the principle of Kirchhoff's current law, according to which the amount of current entering the circuit must be equal to the amount of current leaving the circuit. It continuously monitors the current in the hot wire and neutral wire.
The difference between these two currents is called residual current. When there is an imbalance in the circuit, the residual current will trip the circuit breaker.
The live and neutral wire goes through a zero-sequence current transformer (it is used for sensing an imbalance of current between the two wires). The live and neutral wire is used for current going into and out of the circuit respectively. Since the amount of current is same in both wires, their flux cancels each other.
When the imbalance occurs due to any ground fault, the resultant flux induces a voltage in the current transformer which is connected with a relay that breaks the circuit.



Test
Button
Test Resistor
Phase
Source
Neutral
Relay
Trip
Mechanism
Current ELCB or RCCB
Load
Earth

Type AC RCCB

Such type of RCCB is sensitive to only alternating current and cannot offer protection against DC or any other waveform.

Type A RCCB

Such type of RCCB is sensitive to AC as well as pulsating DC or square waveform

Type B RCCB

Such type of RCCB offers protection against AC up to 1000Hz, Pulsating DC as well as smooth DC.

Limitations of RCCB

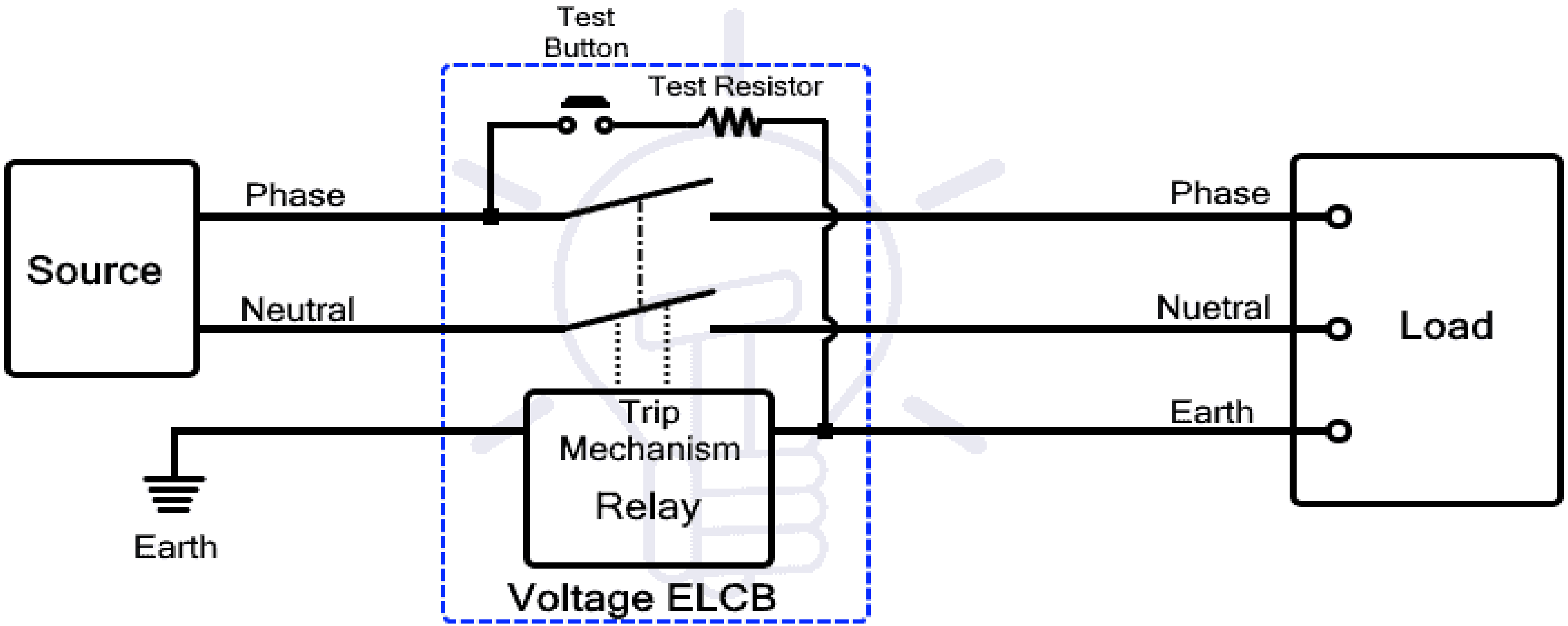
- It does not offer protection against overloading and short circuits.
- It does not protect against line-to-neutral electrical shock.
- It is sensitive to only specific waveforms and does not guarantee protection from other waveforms.

Voltage ELCB

Voltage ELCB operates on voltage level between earth and the body of the equipment. Such ELCB has an extra terminal for earth connection which is directly connected to the load or equipment's body. If the load's body comes in contact with the live wire, it may cause electrical shock upon touching it.

A relay is connected in series with the earthed wire. This relay senses voltage difference between the body and earth. It trips the circuit breaker off if there is a substantial amount of current flow through the earth wire due to the potential difference.

ELCB cannot sense the current leakage if a person comes in contact with a live wire. Therefore, ELCB cannot offer protection for other types of leakage current. Furthermore, it also requires an earth connection, which is not required in RCCB discussed next.



Residual Current Breaker with Overcurrent (RCBO)

RCBO or Residual Current Breaker with overcurrent is a circuit breaker that is made from the combination of RCCB and MCB.

It offers both the functions of the RCCB and MCB i.e. the protection against residual current or ground fault current and overcurrent.

Residual Current: It is the imbalance in the current between live and neutral wire due to the leakage of current to the ground. RCBO offers protection against it to prevent electrical shock.

Overcurrent: Overcurrent means when the current exceeds its limit. It occurs due to two reasons: **Overloading** and **short circuit**.

Overloading: It occurs due to a huge current draw above the rated current for a long duration, which can damage the wiring as well as the components.

Short circuit: It occurs when the live and neutral wire comes into direct contact with each other. There is a huge amount of current flow that can damage electrical equipment.

The RCBO offers protection against both types of fault that is offered individually by RCCB and MCB.